

Côte d'Ivoire Byte-Sized Agriculture Challenge 1st Place Solution

A high-performance agricultural land classification solution using Sentinel-2 satellite imagery, combining deep learning with gradient boosting for multi-temporal crop type identification in Côte d'Ivoire.

Performance Metrics

Public LB Score: **0.973978144**

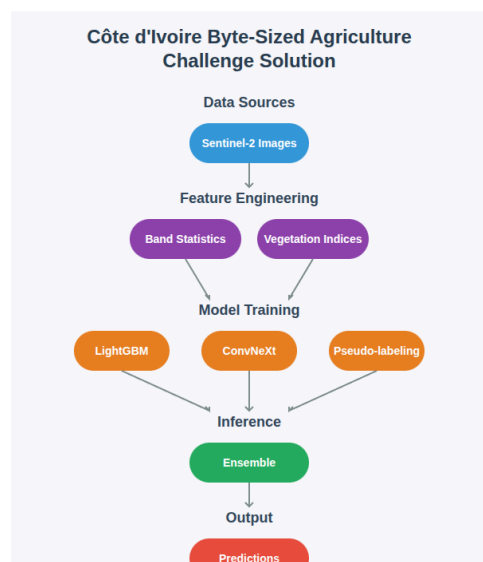
Private LB Score: **0.975995377**

Overview

This solution addresses the Côte d'Ivoire Byte-Sized Agriculture Challenge, requiring classification of agricultural land use patterns using 12-band Sentinel-2 satellite imagery across 12 months. The approach combines ConvNeXt deep learning architecture with LightGBM ensemble methods, optimized for multi-temporal agricultural monitoring.

Architecture

The solution architecture follows this pipeline:



Setup Instructions

Hardware Requirements

- Colab Pro: A100 40GB GPU High RAM option

Model Details

Model Architecture

- **Primary Model:** LightGBM ensemble
- **Secondary Model:** ConvNeXt-Small with multi-modal fusion
- **Input:** 144-channel images (12 bands × 12 months)
- **Classes:** 3 agricultural land use categories
- **Backbone:** ConvNeXt pre-trained on ImageNet

Feature Engineering Pipeline

1. **Spectral Band Statistics:** 8 statistical measures × 12 bands × 12 months = 1,152 features
2. **Vegetation Indices:** 101+ indices (NDVI, EVI, SAVI, etc.) using spyndex library
3. **Temporal Features:** Month-wise aggregations and seasonal patterns

Configuration

Key training parameters:

- **Image Size:** 64×64 pixels (resized from original)
- **Batch Size:** 32 (adjusted for GPU memory)
- **Learning Rate:** 1e-4 with cosine annealing
- **Epochs:** 53 (with early stopping)
- **Validation:** 15-fold stratified cross-validation
- **Augmentations:** Rotation, flip, brightness adjustment

Feature Engineering Details

Spectral Analysis

- **Band Statistics:** Mean, median, min, max, std, skewness, kurtosis, percentiles
- **Temporal Aggregation:** Month-wise and seasonal pattern extraction
- **Normalization:** MinMax scaling with outlier handling

Vegetation Indices

Key indices computed using spyndex library:

- **NDVI:** Normalized Difference Vegetation Index
- **EVI:** Enhanced Vegetation Index
- **SAVI:** Soil-Adjusted Vegetation Index
- **NDWI:** Normalized Difference Water Index
- **101+ additional indices** for comprehensive vegetation analysis



How to Reproduce

1. Upload Brainiac_1stPlace_ByteAgriculture_Solution.ipynb to Colab pro
2. Select A100 40GB GPU with high RAM
3. Run all to get the final_submission file